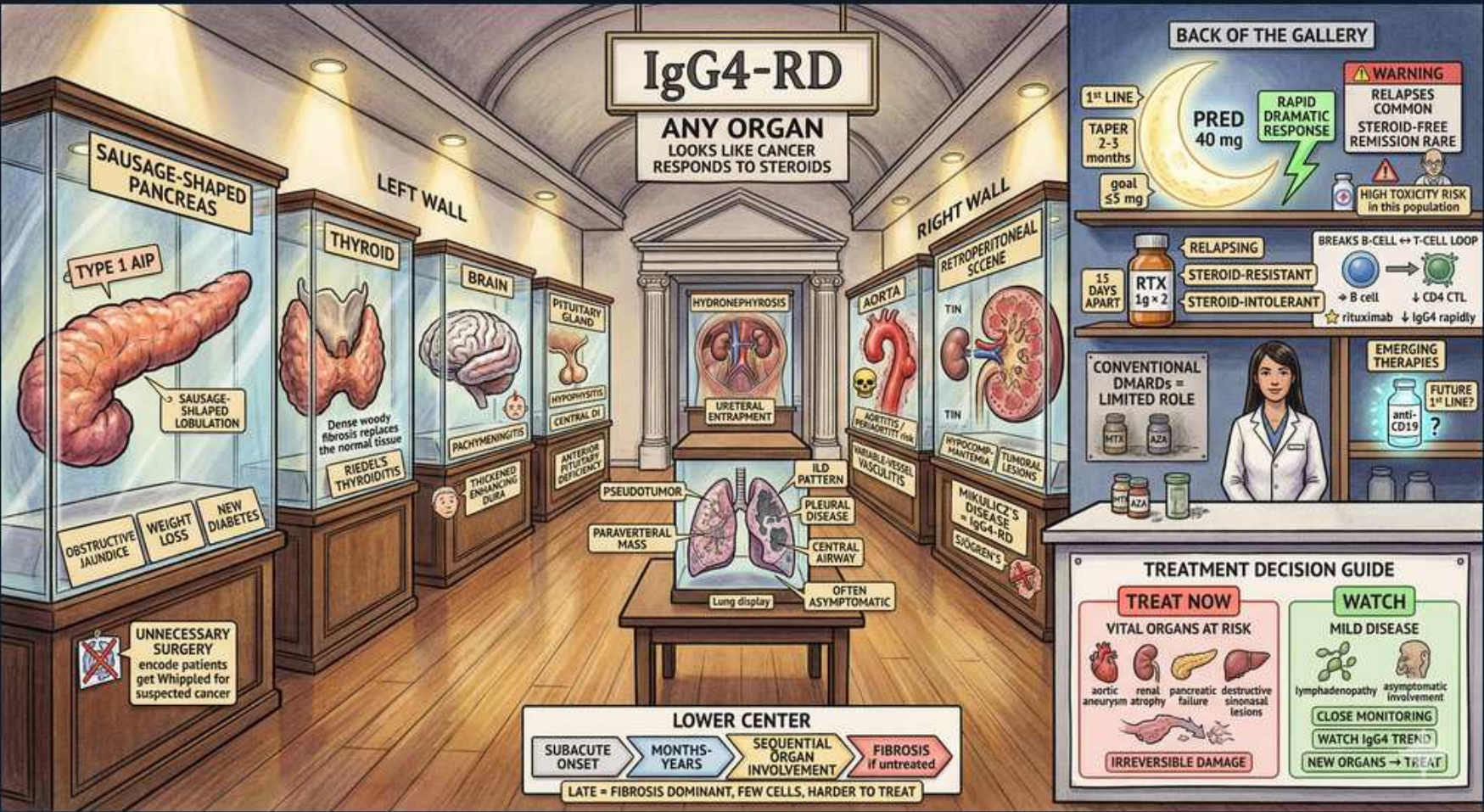


IgG4-Related Disease — The Organ Gallery (Any Organ, Fibrosis & Treatment)



1. Any organ / looks like cancer

WHAT YOU SEE

Header 'IgG4-RD — ANY ORGAN — looks like cancer, responds to steroids'

WHAT IT ENCODES

IgG4-RD is a systemic fibroinflammatory disease that can involve virtually ANY organ — pancreas, biliary tree, salivary/lacrimal glands, orbit, thyroid, kidney, retroperitoneum, aorta, lung, pituitary, lymph nodes, meninges — typically forming mass lesions or diffuse organ enlargement that mimic tumors. The shared histology across sites is storiform fibrosis + IgG4+ lymphoplasmacytic infiltrate + obliterative phlebitis. Hallmark unifying behavior: tumor-like masses that respond dramatically to corticosteroids.

BOARD TRAP

WRONG answer: working up each organ mass as a separate primary malignancy. Discriminator: when an older man develops sequential or synchronous steroid-responsive 'tumors' across multiple organs with elevated serum IgG4, think a single unifying diagnosis — IgG4-RD — and biopsy the most accessible site rather than pursuing multiple cancer resections.

2. Sausage-shaped pancreas (Type 1 AIP)

WHAT YOU SEE

Left wall pancreas, 'TYPE 1 AIP', sausage lobulation, new diabetes

WHAT IT ENCODES

Type 1 autoimmune pancreatitis is the prototypical IgG4 organ lesion: a diffusely enlarged, sausage-shaped pancreas with a hypoattenuating capsule-like rim and irregular main-duct narrowing, presenting with painless obstructive jaundice, weight loss, and frequently new-onset diabetes from islet involvement. Serum IgG4 is often high; duct involvement may cause IgG4-related sclerosing cholangitis. It melts with steroids.

BOARD TRAP

WRONG answer: proceeding to Whipple resection for presumed pancreatic adenocarcinoma. Discriminator: Type 1 AIP shows a DIFFUSE sausage gland with capsule rim and duct narrowing (not a focal mass with upstream duct dilation), elevated IgG4, other-organ disease, and steroid responsiveness — confirm with EUS-FNA/biopsy and a steroid trial before resecting, since AIP is the great pancreatic-cancer mimic.

3. Thyroid (Riedel / fibrosing)

WHAT YOU SEE

Thyroid panel, fibrosing thyroiditis

WHAT IT ENCODES

Riedel thyroiditis (invasive fibrosing thyroiditis) is part of the IgG4 spectrum — the gland becomes rock-hard and woody, with dense fibrosis extending beyond the capsule into surrounding neck structures, causing compressive symptoms (dysphagia, stridor, hoarseness) and often hypothyroidism. The IgG4-related fibrosing variant of Hashimoto thyroiditis is also recognized.

BOARD TRAP

WRONG answer: diagnosing anaplastic thyroid carcinoma based on a fixed, stony-hard, rapidly enlarging thyroid mass. Discriminator: anaplastic carcinoma occurs in the elderly with aggressive growth and a poor prognosis on FNA/biopsy showing pleomorphic malignant cells; Riedel thyroiditis shows bland dense fibrosis with IgG4+ plasma cells, is steroid-responsive, and often involves other IgG4 organs — open biopsy distinguishes them.

4. Brain — hypophysitis / pachymeningitis

WHAT YOU SEE

Brain panel, hypophysitis, pituitary

WHAT IT ENCODES

IgG4 hypophysitis causes pituitary stalk thickening and gland enlargement with anterior hypopituitarism and/or posterior diabetes insipidus (polyuria, polydipsia, hypernatremia). IgG4-related hypertrophic pachymeningitis causes dural thickening with headaches, cranial neuropathies, and mass effect. Both respond to glucocorticoids, often with improvement in hormone deficits if treated early.

BOARD TRAP

WRONG answer: calling a sellar mass with hormone deficiency a pituitary adenoma and planning transsphenoidal surgery. Discriminator: a pituitary adenoma rarely causes diabetes insipidus and grows as a discrete adenomatous mass, whereas IgG4 hypophysitis shows stalk thickening, DI, other-organ IgG4 involvement, and steroid responsiveness — recognize it to avoid unnecessary neurosurgery.

5. Kidney — tubulointerstitial nephritis

WHAT YOU SEE

Kidney panel, TIN, low complement

WHAT IT ENCODES

The dominant renal lesion of IgG4-RD is IgG4-related tubulointerstitial nephritis (TIN) — patchy, often radiologically apparent cortical lesions or masses, with a dense IgG4+ plasma-cell interstitial infiltrate and tubular basement membrane immune deposits. Hypocomplementemia (low C3/C4) is common and signals active disease. Membranous nephropathy can also occur. It responds to steroids.

BOARD TRAP

WRONG answer: attributing renal masses + low complement to lupus nephritis or a renal malignancy. Discriminator: IgG4 renal disease shows TIN with IgG4+ plasma cells and TBM deposits (not the immune-complex glomerulonephritis of lupus, which has positive ANA/dsDNA and 'full-house' immunofluorescence); biopsy with IgG4 staining plus extrarenal IgG4 organ involvement clinches it.

6. Retroperitoneal fibrosis

WHAT YOU SEE

Right wall 'RETROPERITONEAL FIBROSIS', ureteral entrapment

WHAT IT ENCODES

Retroperitoneal fibrosis (Ormond disease, often IgG4-related) is a fibrous plaque encasing the abdominal aorta, IVC, and ureters. It is frequently SILENT until it entraps the ureters, causing hydronephrosis and obstructive (post-renal) acute kidney injury, or causes vague back/flank/abdominal pain and venous obstruction. Imaging (CT/MRI/PET) shows a periaortic soft-tissue rind; it is steroid-responsive, and ureteral stents may be needed acutely.

BOARD TRAP

WRONG answer: working up rising creatinine as intrinsic renal disease and missing the obstruction. Discriminator: IgG4 retroperitoneal fibrosis silently entraps ureters → bilateral hydronephrosis and obstructive renal failure; image the retroperitoneum (and relieve obstruction with stents/nephrostomy) before it becomes irreversible, then treat the fibrosis with steroids ± rituximab.

7. Periaortitis / aortitis

WHAT YOU SEE

Aorta panel, periaortitis, aneurysm risk

WHAT IT ENCODES

Chronic periaortitis/aortitis from IgG4-RD causes circumferential aortic wall thickening (a periaortic 'cuff'), most often in the infrarenal abdominal aorta, and is a recognized cause of inflammatory (non-infectious, non-atherosclerotic) aortic aneurysm and dissection risk. It frequently coexists with retroperitoneal fibrosis. Recognizing it allows immunosuppression and aneurysm surveillance.

BOARD TRAP

WRONG answer: labeling all aortic wall inflammation as giant cell or Takayasu arteritis. Discriminator: GCA/Takayasu are granulomatous large-vessel vasculitides (elevated ESR/CRP, claudication, age/sex patterns) typically without IgG4+ plasma-cell-rich storiform fibrosis; IgG4 periaortitis shows the characteristic histology, bland CRP, periaortic soft tissue, and association with other IgG4 organs.

8. Lung — pleural / pseudotumor

WHAT YOU SEE

Central lung panel, pleural mass, pseudotumor

WHAT IT ENCODES

Pulmonary IgG4-RD is protean: solid nodules/masses (inflammatory pseudotumor), peribronchovascular and septal thickening, ground-glass opacities, interstitial fibrosis, hilar/mediastinal lymphadenopathy, and pleural thickening/effusions. Many patients are asymptomatic with incidental findings; symptoms include cough and dyspnea. Lesions respond to steroids.

BOARD TRAP

WRONG answer: assuming a solitary lung nodule/mass plus mediastinal nodes is bronchogenic carcinoma or sarcoidosis without tissue. Discriminator: IgG4 pulmonary pseudotumor responds to steroids and shows IgG4+ plasma cells with storiform fibrosis on biopsy; sarcoid shows non-caseating granulomas and cancer shows malignant cells — biopsy is required because imaging alone cannot separate IgG4 pseudotumor from malignancy.

9. Subacute, fibrosis-dominant late

WHAT YOU SEE

Lower banner: subacute onset, months–years, fibrosis if untreated

WHAT IT ENCODES

IgG4-RD runs a subacute, indolent course over months to years, with sequential organ involvement and tumor-like lesions; it is rarely acutely life-threatening but causes cumulative organ damage. Early disease is inflammation-rich (plasma cells, plasmablasts) and very steroid-responsive; untreated late disease becomes fibrosis-DOMINANT with few plasma cells, making it steroid-refractory and producing irreversible organ failure.

BOARD TRAP

WRONG answer: adopting a 'watch and wait' approach indefinitely because the disease is indolent. Discriminator: late fibrotic lesions have sparse plasma cells and respond poorly to immunosuppression, so the window for effective treatment is the inflammatory phase — treat before irreversible fibrosis sets in, particularly when vital organs are involved.

10. Prednisone first-line

WHAT YOU SEE

Top-right 'PRED 40 mg', rapid dramatic response, taper

WHAT IT ENCODES

Glucocorticoids are first-line induction therapy for active IgG4-RD: typically prednisone ~0.6 mg/kg/day (often around 30–40 mg/day) for 2–4 weeks, then a gradual taper over months. The response is rapid and dramatic — masses shrink, organ function recovers, and serum IgG4 falls — which is itself supportive of the diagnosis.

BOARD TRAP

WRONG answer: continuing to call it IgG4-RD when there is NO steroid response. Discriminator: steroid responsiveness is so characteristic that failure to respond should prompt re-biopsy and reconsideration of malignancy (lymphoma, carcinoma) or another mimic — a non-responding 'IgG4 mass' is a red flag for missed cancer, not just refractory disease.

11. Relapses common → rituximab

WHAT YOU SEE

RTX panel, 'relapses common', B-cell depletion, CD20

WHAT IT ENCODES

Relapse is common as steroids are tapered or stopped. Rituximab — an anti-CD20 monoclonal antibody that depletes B cells and plasmablasts — is the key steroid-sparing agent for relapsing, refractory, or steroid-intolerant disease, and is increasingly used for induction in severe multi-organ involvement. Plasmablast counts can guide and predict relapse. Conventional DMARDs (azathioprine, mycophenolate, methotrexate) have only a limited, adjunctive role.

BOARD TRAP

WRONG answer: reaching for a conventional DMARD (e.g., methotrexate or azathioprine) as the primary steroid-sparing/relapse therapy. Discriminator: in IgG4-RD the disease is B-cell/plasmablast-driven, and RITUXIMAB (anti-CD20) is the evidence-favored go-to for relapsing or refractory disease — conventional DMARDs are weakly effective and not the preferred answer on boards.

12. Treat now vs watch

WHAT YOU SEE

'Treatment Decision Guide': TREAT NOW (vital organs at risk) vs WATCH (mild)

WHAT IT ENCODES

Treatment urgency is risk-stratified. TREAT NOW when vital/critical organs are threatened — aortitis with aneurysm risk, retroperitoneal fibrosis entrapping ureters, autoimmune pancreatitis/cholangitis with biliary obstruction, tubulointerstitial nephritis, hypophysitis, pachymeningitis. WATCH (with close monitoring) for mild, non-organ-threatening disease such as isolated lymphadenopathy or asymptomatic single-gland involvement. The threshold to treat is the threat of irreversible organ damage.

BOARD TRAP

WRONG answer: 'watching' a patient whose disease threatens a vital organ (e.g., ureteral entrapment, biliary obstruction, aortic involvement). Discriminator: irreversible organ damage is the trigger to treat NOW — observation is only appropriate for mild, non-threatening disease; allowing threatened vital organs to progress to fibrosis is a management error because late fibrotic disease is steroid-refractory.